# Paleomagnetic Study of a Vertical Sequence of Deccan Traps from Jabalpur

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## Abstract

The palaeomagnetism of a vertical sequence of Deccan traps ranging in elevation from 1750 to 2900 ft. along the eastern margins of the Deccan trap exposures near Jabalpur, has been studied. Eleven different flows were sampled at approximately 100 ft. elevation interval. Nine flows at elevations between 2150 and 2900 ft. showed normal polarization with up dip. The mean direction for these siving unit weight to each flow was, $D = 343°$E and $I = 28°$(up). Two flows at elevations between 1740 and 2000 ft. showed intermediate directions with down dip. It appears that the latter corresponds to a zone of actual field reversal which has been encountered at several locations on the territory of the Deccan traps. The mean directions of the nine upper flows give a reliable estimate of the palaeosecular variation of the geomagnetic field during the period of normal polarity. The mean directions obtained from vertical sequences of traps studied from different localities are compared for estimating the amount of continental drift that might have taken place during the period of main Deccan trap activity. Correlation of magnetic directions of flows from various localities indicates that the geomagnetic field reversal at about 2000 ft. elevation can be traced over a large territory of the traps.

## Introduction

Significant contributions to the study of the palaeomagnetion of the Deccan Traps have ben made by CLEGG *et al.* (1956), DEUTSCH *et al.* (1958), SAHASRABUDHE (1963), VERMA and RAO (1963), BHIMASANKARAM and PAUL (1968). The studies made so far have revealed a pronounced

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geomagnetic field reversal in the Deccan traps which is believed to be located near an elevation of about 2000 ft. above mean sea level (DEUTSCH *et al.*, 1959). The rocks above this elevation have been found to be generally « normally » magnetised (*i.e.* in the present sense of the Earth's field) and those at lower elevations are found to have « reverse » polarity. Recently BHIMSANKARAM and PAL (1968) have studied a vertical sequences of traps ranging in elevation from 1670

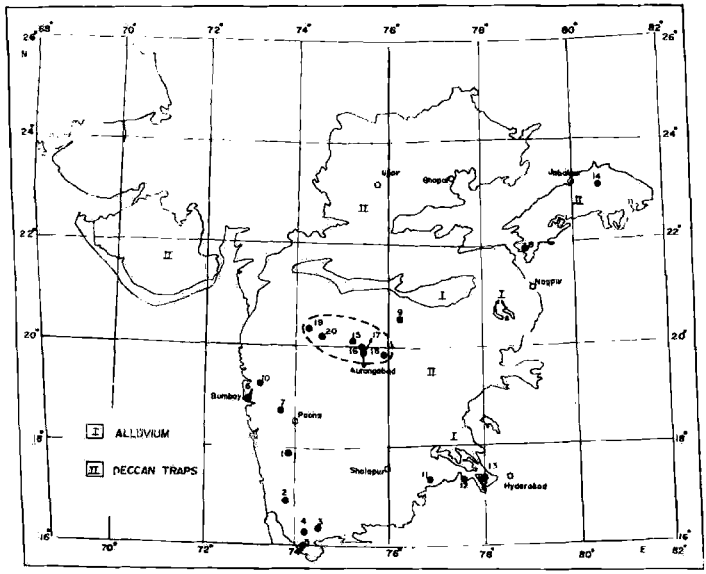



FIG. 1 - Map showing the locations of sites from where rocks of the Deccan Traps have been studied so far. 1: Mahableshwar; 2: Amba; 3: Nipani; 4: Gargoti; 5: Ajra; 6: Bombay; 7: Khandala; 8: Linga; 9: Buldana; 10: Igatpuri; 11: Gulbarga; 12: Chincholi; 13: Vikarabad; 14: Jabalpur; 15: Ellora; 16: Harsul; 17: Aurangabad; 18: Jalna; 19: Chandor; 20: Ankai.

ft. to 3000 ft. from several localities including Ellora, Aurangabad, Chandor, Ankai, Hursul and Jalna (Fig. 1) situated in the northwestern margins of the main exposures. These studies have indicated the possibile existence of more than one reversal within the Deccan traps. SAHASRABUDHE (1963) has reported results of studies of 27 different flows from Mahabaleshwar (17°55′ N; 73°38′ E), 5 flows each from Nipani (16°26′ N; 74°22′ E) and Amba (16°59′ N; 73°46′ E) and 16 flows from Khandala (17°55′ N; 73°38′ E). The results of the studies from Mahableshwar show a high degree of scatter. Fig. 1 shows the localities where the Deccan traps have been studied so far.

In the present paper results of palaeomagnetic studies of a vertical sequence of traps from the eastern margins near Jabalpur are discussed. The main object of these studies has been to obtain a clear picture of the secular variation of the geomagnetic field during the period of eruption of these traps, to learn about the nature of the geomagnetic field during the period of field reversal and to correlate Deccan traps from different areas as far as we can.

## Geologic Setting

The Deccan traps overlie Lameta beds near Jabalpur, fossil evidence for which suggests a Turonian (Middle Cretaceous) age according to KRISHNAN (1968). In Narmada Valley the traps are underlain by Bagh beds (Cenomian to Senonian age). For some of the Intertrappean beds the age of Eocene has been suggested from fossil evidence by Sahni (quoted by KRISHNAN, 1968, p. 418). It has been generally concluded that the Deccan traps commenced to pour out during Upper Cretaceous times and continued till well into Eocene times. Paleomagnetic studies offer a very useful tool for studying the history of eruption of the Deccan traps over their vast territory. On the basis of paleomagnetic parameters it is possible to correlate the flows over large distance of the order of a few hundred kilometers.

## Sampling

About one hundred oriented hand samples were collected from natural outcrops of Deccan trap exposures between lat. 23°03′ N to 23°12′ N and long. 80°31′ E to 81°58′ E. From each outcrop, eight to ten samples were obtained to satisfy the statistical requirement. The elevations of the sites were determined from detailed topographic sheets (1″ = 1 mile) and ranged from 1750 ft. to 2900 ft. The samples were cored into cylindrical specimens and studied in the laboratory with astatic magnetometers and other conventional instruments used for removing unstable components of magnetization.

## NRM Measurements

The results of NRM measurements are shown in Figs. 2 and 3. It is seen that the scatter in NRM directions was different at different sites. The mean intensity of magnetization for various flows varied from $2.2 \times 10^{-3}$ to $12.3 \times 10^{-3}$ gauss. The total range in magnetic

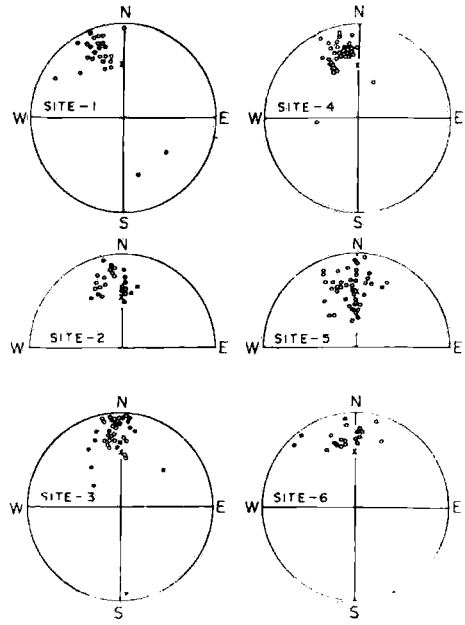



FIG. 2 - NRM directions of sites 1 to 6 plotted on stereographic projections. North seeking pole, down - ● ; up - ○.

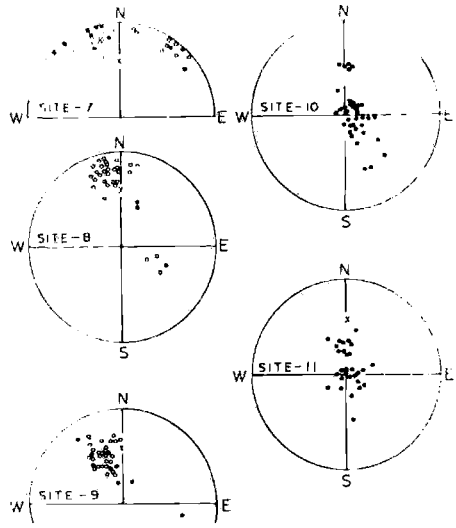



FIG. 3 - NRM directions of sites 7 to 11 plotted on stereo graphic projections. North seeking pole, down - ● ; up - ○.

intensity was $1.1 \times 10^{-3}$ to $24.7 \times 10^{-3}$ gauss. The mean values of intensity for each flow are plotted in Fig. 7. The variations in magnetic intensity ($J_n$) suscephibility ($K$) and $Q_n$ ratio are quite appreciable for each site, the maximum value of $K$ being about three times the minimum. A wide variation in NRM intensity from flow to flow is quite obvious. Although the majority of specimens from sites 1 to 9 show normal polarity and up dip, there were few specimens from several sites which had normal polarity with down dips. Samples from sites 10 and 11 showed intermediate direction and steep down dips, completely different from those of sites 1 to 9.

## Stability Tests

It was necessary to use stability tests in order to remove the viscous components of magnetization acquired in the earth's field

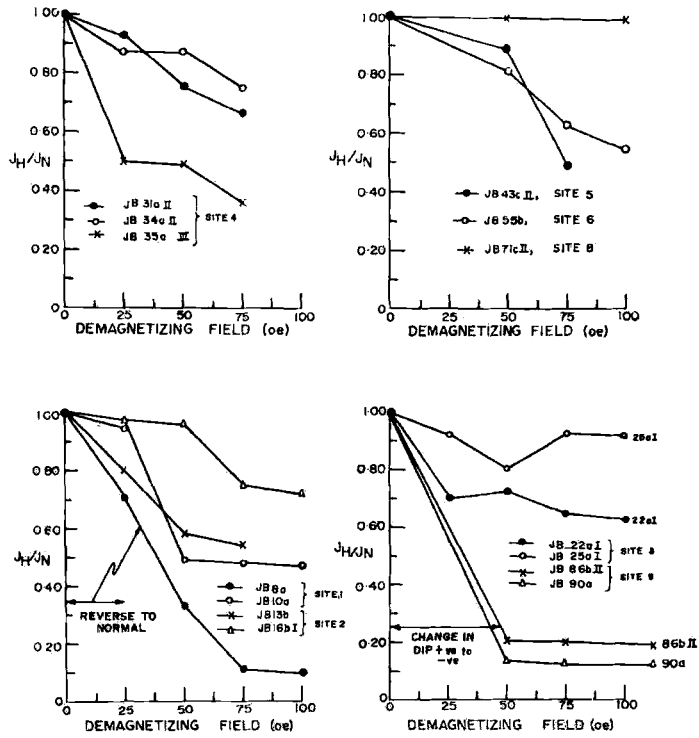



FIG. 4 - A.C. demagnetization curves of normalized intensity for individual specimens, in a few interesting cases.

subsequent to the formation of the rocks. For this purpose a few pilot specimens from each site were subjected to alternating field

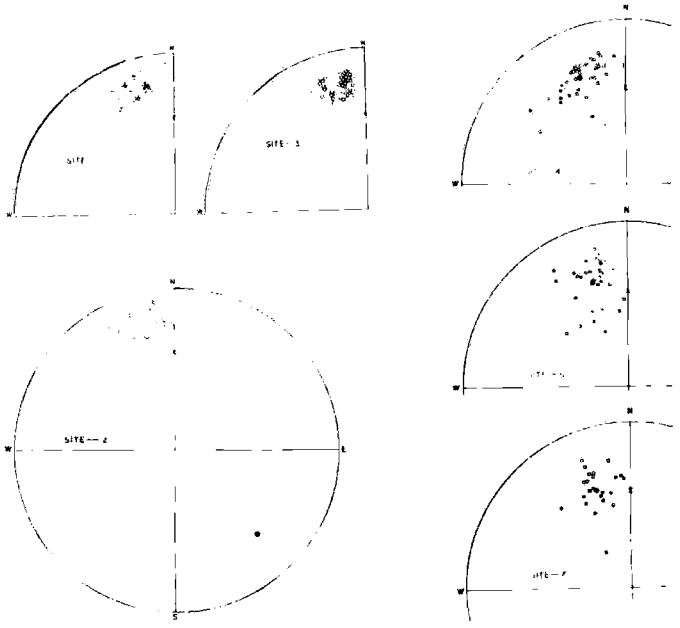


FIG. 5 - Directions of magnetization for specimens from sites 1-6 after a.c. cleaning.

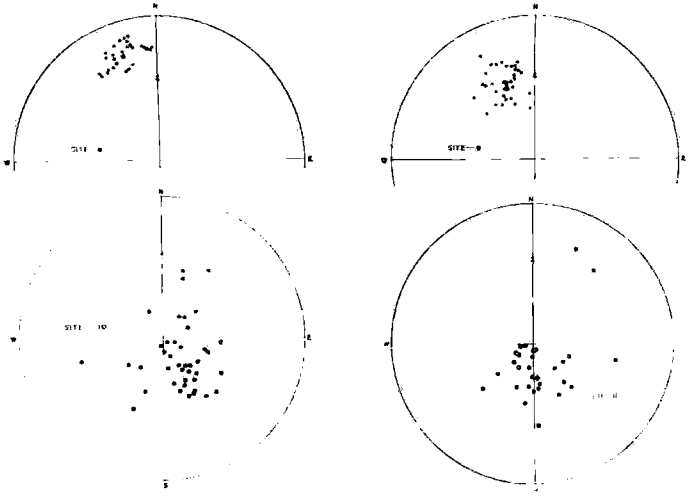


FIG. 6 - Directions of magnetization for specimens from sites 8-11 after a.c. cleaning.

demagnetization in progressively increasing fields of 25, 50 and 100 oe peak field in an apparatus similar to that used by CREER (1959). The behaviour of the samples from different sites was quite different. An optimum field of 50 oe was found to be sufficient for cleaning most of the remaining specimens; however, it was found necessary to clean some of the specimens in 100 oe peak field to remove unstable components.

A few typical a.c. demagnetization curves of normalised intensity for specimens from different sites are shown in Fig. 4. In most of the cases an appreciable change in declination and inclination took place in fields up to 50 oe. In some cases reversely magnetised specimens changed to normal as a result of cleaning. There were always a few abnormal specimens from different sites which showed increase in intensity upon a.c. cleaning rather than a decrease as is normally the case. These specimens suggest that there were at least two opposing components of magnetization in the specimens, the demagnetization of one of which in low fields resulted in an increase of intensity. Appreciable increase in the precision parameter k after cleaning was noticed in the case of all the sites. Directions of magnetization after cleaning are shown in Figs. 5 and 6.

## Results and Discussion

*General Properties.*

The mean directions of magnetization before and after a.c. cleaning are given in Table 1 for various flows, along with other paleomagnetic parameters. Mean declination and inclination (after cleaning), susceptibility, $Q_n$ ratio, $J_n$ and $J_h$ are plotted in Fig. 7. It is interesting to note a gradual, though not smooth, increase in inclination with increasing age of the flows down to 2150 ft. The corresponding change in declination is very small. At an elevation of about 2000 ft., the inclination as well as declination show an abrupt change. Flows No. 10 and 11 show intermediate directions of magnetizations as compared to the normal ones from 1 to 9. This is close to the elevation near which the geomagnetic field reversal has been previously noted by DEUTSCH *et al.* (1959) along the west coast of India. The magnetic susceptibility values were determined in a field of 0.5 oe for all specimens from the various flows. The mean values (plotted in Fig. 7) do not show a clear correlation with NRM intensity. The

TABLE 1 - Mean directions of magnetization before and after a.c. cleaning. Paleomagnetic parameters for sites 1 to 11, Jabalpur.

| Site No. | Location N | Location E | Elevation (ft) above mean sea level | N | n | Before Cleaning Mean $D^\circ$ | Before Cleaning Mean $I^\circ$ | Before Cleaning $R$ | Before Cleaning $k$ | Before Cleaning $\alpha_{95}$ | After Magnetic Cleaning $D^\circ$ | After Magnetic Cleaning $I^\circ$ | After Magnetic Cleaning $R$ | After Magnetic Cleaning $k$ | After Magnetic Cleaning $\alpha_{95}$ | After Magnetic Cleaning $\lambda$(°S) | Virtual Geomagnetic Pole $\lambda_p$(°N) | Virtual Geomagnetic Pole $L_p$(°W) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | 23°04′ | 80°57′ | 2900 | 8 | 27 | 339 | −11 | 5.69 | 3 | 29 | 343 | −19 | 7.98 | 350 | 3 | 10 | 53 | 70 |
| 2. | 23°11′ | 81°00′ | 2750 | 10 | 25 | 351 | − 7 | 9.36 | 14 | 12 | 347 | −19 | 8.95 | 160 | 4 | 10 | 54 | 77 |
| 8. | 23°06′ | 80°47′ | 2650 | 9 | 35 | 357 | −14 | 7.02 | 4 | 23 | 342 | −24 | 6.93 | 86 | 6 | 13 | 50 | 71 |
| 3. | 23°12′ | 81°01′ | 2600 | 9 | 41 | 357 | − 3 | 8.57 | 19 | 11 | 348 | −23 | 8.97 | 267 | 3 | 12 | 53 | 80 |
| 5. | 23°03′ | 80°59′ | 2450 | 9 | 43 | 358 | −20 | 8.28 | 11 | 14 | 342 | −33 | 8.84 | 50 | 7 | 18 | 45 | 74 |
| 4. | 23°08′ | 81°01′ | 2350 | 10 | 38 | 343 | −19 | 9.88 | 75 | 5 | 337 | −28 | 9.90 | 90 | 5 | 15 | 46 | 66 |
| 7. | 23°06′ | 80°49′ | 2300 | 7 | 28 | 359 | −12 | 6.48 | 12 | 15 | — | — | — | — | — | — | — | — |
| 6. | 23°03′ | 80°58′ | 2200 | 9 | 23 | 350 | −11 | 8.68 | 22 | 10 | 340 | −36 | 8.90 | 89 | 5 | 20 | 43 | 73 |
| 9. | 23°09′ | 80°45′ | 2150 | 10 | 36 | 337 | −19 | 8.99 | 9 | 15 | 338 | −41 | 8.90 | 76 | 5 | 24 | 38 | 73 |
| 10. | 23°12′ | 80°31′ | 2000 | 10 | 41 | 101 | +78 | 9.17 | 14 | 12 | 148 | +71 | 9.65 | 26 | 9 | 56 | 6 | 82 |
| 11. | 23°12′ | 80°32′ | 1740 | 9 | 30 | 311 | +70 | 8.37 | 13 | 13 | 176 | +70 | 8.87 | 27 | 9 | 5 | 13 | 97 |
| Mean for sites 1 to 9 | 23° 8′ | 80°51′ | — | — | — | — | — | — | — | — | 343 | −28 | 7.92 | 87.5 | 5 | 15 | 48 | 74 |

N and n = No. of samples and specimens for NRM studies.
$D$ = Declination, degrees clockwise from the geographic north, and $I$ = Inclination, degrees +ve (down), −ve (up).
$R$ = Length of the resultant of N vectors; $k = (N-1/N-R)$, Fisher's precision parameters.
$\alpha_{95}$ = Circle of confidence at 95 % probability level.
$\lambda$ = Ancient latitude of the site.
$\lambda_p$ and $L_p$ = Latitude and longitude of the virtual geomagnetic pole (VGP).

$Q_n$ ratios determined from the susceptibility values (using a values of 0.50 oe for intensity of earth's field) show a fairly good correlation with NRM intensities. Mean $Q_n$ ratios show a variation from 2.4 to about 20, while the total range of $Q_n$ values for all specimens is 1 to 117. This range seems to be characteristic of Cenozoic and slightly older basaltic flows and indicates wide differences in the nature of magnetization of the specimens.

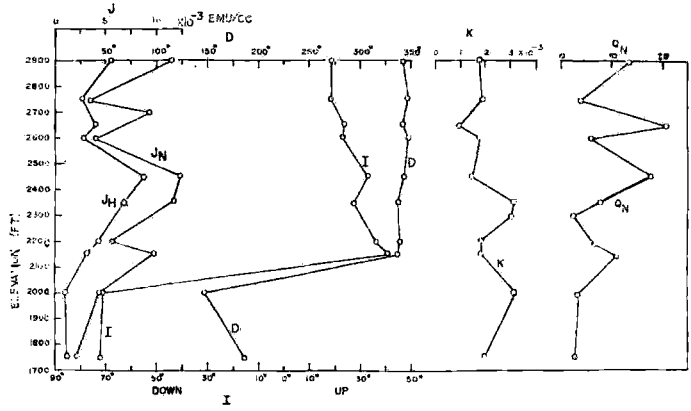



FIG. 7 - Palaeomagnetic parameters for various sites; $J_N$ = intensity of NRM; $J_H$ = intensity after a.c. cleaning; $D$ = declination (east of north); $I$ = inclination (−ve, up); $K$ = magnetic susceptibility (c.g.s. units); $Q_N$ = Koenigsberger ratio.

*Secular Variation.*

The mean direction of magnetization for each flow after a.c. cleaning may be taken as a representative direction of the field at the time when the basaltic magma cooled below the Curie-temperature. We therefore have a fairly good record of the behaviour of the geomagnetic field during the period of eruption of flows 1 to 11. The virtual geomagnetic poles corresponding to the mean direction of each flow are shown in Fig. 8. The mean pole position for flows 1 to 9 giving unit weight to each mean direction is also shown in this figure. The arrows in the figure point towards the direction of the younger flows. It appears that the variation of the pole positions from flows 9 to 1 (about 15° in latitude) could be ascribed to polar wandering or secular variation of the geomagnetic field during the period of their extrusion. It is unlikely that such large difference in pole position could be due to continental drift during the period of

their eruption. There is, however, no clear indication of a clockwise rotation of the main field about the mean position. This may be due to an insufficient number of flows having been sampled in the sequence. The large change in the pole position indicated by flows 10 and 11 reflects an abrupt change in the nature of the geomagnetic field. It appears most likely that flows with reverse polarity are underlying flows 10 and 11, (this has been confirmed subsequently).

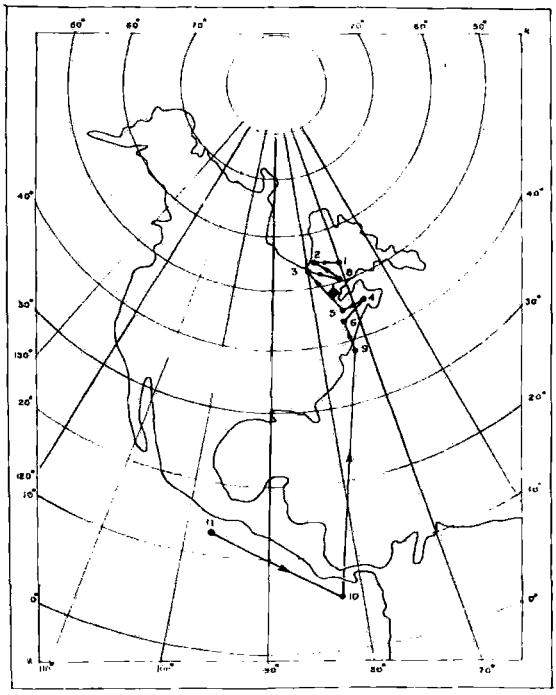


FIG. 8 - Virtual geomagnetic poles corresponding to each flow of the vertical sequence. ◆ represent mean direction for the normal flows in the vertical sequence studied.

If this is the case, then flows 10 and 11 are representative of the field during the period of reversal and should give us some idea about the nature of the field during this period.

A reliable estimate of the amount of secular variation of the geomagnetic field is essential for interpreting the palaeomagnetic results in terms of continental drift of the landmasses.

It has been generally argued that the site mean directions particularly in igneous rocks approximate to spot readings of the geo-

TABLE 2 - Paleomagnetic parameters of vertical sequences of flows from five localities in the Deccan traps.

| No. | Locality | Elevation (ft) | No. of flows or sites | Mean Direction D | I | R | k | $\alpha_{95}$ | $\delta$ | Virtual Geomagnetic Pole $\lambda$(°S | $\lambda_p$(°N) | $L_p$(°W) | dp | dm | polarity | reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | Mahableshwar 17°55′N, 73°38′E | 2000-3500 | 5 flows | 345 | −40 | 4.90 | 40 | 10 | 11° | 23 | 47 | 86 | 7 | 12 | N | SAHASRABUDHE (1963) |
| 2. | Mahableshwar 17°55′N, 73°38′E | 200-2000 | 15 flows | 153 | +52 | 13.80 | 12 | 11 | 23° | 33 | 33 | 80 | 10 | 15 | R | » |
| 3. | Amba 16°59′N, 73°46′E | 500-1800 | 5 flows | 151 | +56 | 4.96 | 100 | 6 | 7° | 37 | 30 | 80 | 6 | 9 | R | » |
| 4. | Khandala 18°45′N, 73°22′E | 400-2000 | 15 flows | 147 | +58 | 14.70 | 47 | 5 | 11° | 39 | 25 | 79 | 5 | 7 | R | » |
| 5. | Jabalpur 23° 4′N, 80°57′E | 2150-2900 | 8 flows | 343 | −28 | 7.92 | 87.5 | 5 | 8° | 15 | 48 | 74 | 3 | 6 | N | This paper |

*D* = Declination in degrees clockwise from geographic north, and *I* = inclination in degrees, +ve down and −ve up.

*R* = Length of the resultant vector, and $k$ (=N−1/N−R) precision parameter.

$\alpha_{95}$ = 95 % circle of confidence, and $\delta$ = angular dispersion factor.

$\lambda$ = Ancient latitude of the site.

$\lambda_p$ and $L_p$ = Latitude and longitude, respectively, of the virtual geomagnetic pole (VGP).

*dp* and *dm* = Semi-axes of the elliptical error area round the pole at 95 % probability.

*N* = Normal polarity, and R = reverse polarity.

magnetic field, so that the scatter observed in the site mean directions would reflect the magnitude of secular variation during the times in which the rocks of the various sites belonging to the same rock unit were laid down. We have data for site mean directions from several vertical sequences in the Deccan traps comprising 6 to 20 flows from the Mahableshwar, Nipani, Khandala, Amba and Jabalpur areas. As suggested by CREER (1965) secular variation can be estimated from the angular dispersion factor $\hat{\delta} = \cos^{-1} \frac{R}{N}$ or circular standard deviation $\Delta = \frac{81}{\sqrt{k}}$ calculated for these flows. Here $R$ is the resultant of $N$ unit vectors and $k$ is Fisher's precision parameter. The angular

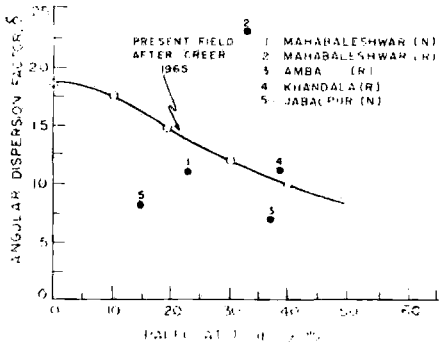



FIG. 9 - Values of the angular dispersion factor $\hat{\delta}$ vs. palaeolatitude for various sequences of flows.

dispersion factor for each vertical sequence studied has been calculated from the mean direction for that sequence. The paleomagnetic data from various localities including the mean direction, precision parameter ($k$), ancient latitude of site ($\lambda$), pole position (V.G.P.) and angular dispersion factor ($\hat{\delta}$) are given in Table 2. The values of $\hat{\delta}$ are plotted as palaeolatitudes of sites (in °S) in Fig. 9. CREER'S (1965) estimate of dispersion for the present best fitting dipole field is also plotted in the figure. It is seen that with the exception of data from the Mahabaleshwar traps, the angular dispersion factor corresponding to various sequences studied is less than or equal to the angular dispersion expected from the present geomagnetic dipole field.

These results give a strong indication that the geomagnetic field during the Deccan trap period (considered to be mostly Upper Cretaceous to Eocene) must have been essentially of dipole nature.

*Palaeointensity During Field Reversal.*

During the period of geomagnetic field reversal, a decrease in the intensity of the geomagnetic field has been reported by several investigators. From Icelandic lavas, SIGURGEIRSSON (1957) has estimated the intensity of the revesring field to be about 1/3 of the initial intensity. ZIJL *et al.* (1962) have estimated a decrease of 20 to 25 % in intensity

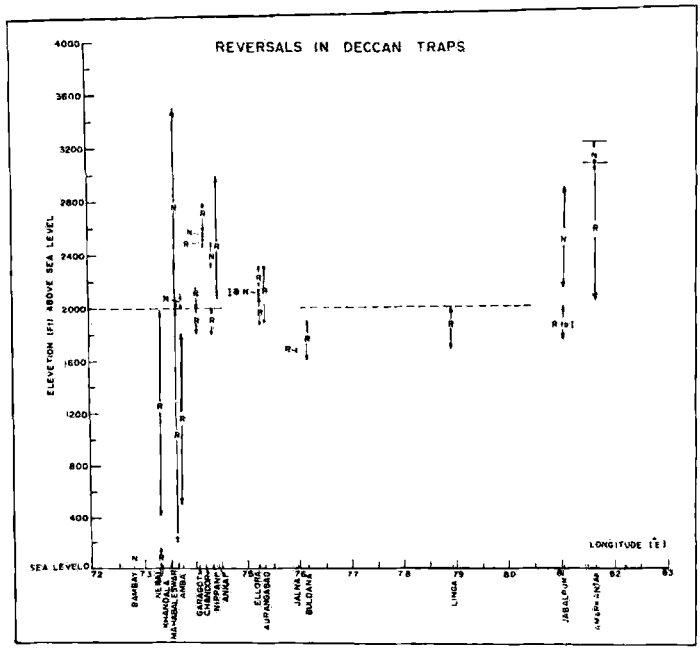



FIG. 10 - Correlation of Deccan Traps polarities for various localities.

of the field during reversal observed in the Stormberg lavas of Late Triassic to Early Jurassic age in South Africa. For the present field reversal investigated, no reliable estimate of decrease in field intensity has been made yet. It is worth mentioning, however, that the intensity $J_h$ of the flows after magnetic cleaning does show minimum values for flows 10 and 11 which are interpreted to represent the period of reversal. On the other hand, there seems to be no systematic difference in magnetic susceptibility between the latter flows and those lying them above.

*Correlation of Reversals in the Traps.*

It is interesting to see if we could find any correlation within the Deccan traps on the basis of their directions of magnetization. The Deccan traps investigated so far extend laterally from longitude nearly 71° E to 81° E. An attempted correlation of their vertical sequences is shown in Fig. 10.

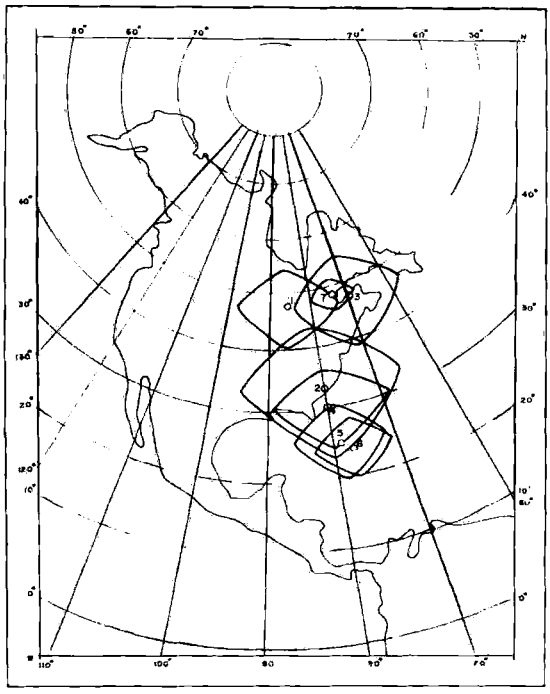


FIG. 11 - Pole positions corresponding to vertical sequences of Traps from various localities. 1: Mahableshwar (N); 2: Mahableshwar (R); 3: Amba (R); 4: Khandala (R); 5: Jabalpur (N).

This figure suggests that « reverse » rocks are found mostly at elevations below 2000 ft. over a large territory, and the « normal » rocks occur at higher elevations. The normal rocks found near Bombay have been interpreted as being due to faulting near the west coast. The one conspicuous « anomalous » area found so far is near Aurangabad, Harsul and Ellora where rocks with reverse polarity have been found up to about 3000 ft. elevation. There are two alternative explanations for these anomalous polarities —1. This area has had

an independent history of magmatic eruptions, so that volcanisum during the period of reverse polarity continued here while it had ceased at other places; or 2. The area has suffered a differential uplift of the order of 1000 ft. with respect to the adjoining areas after the period of eruption. More work needs to be done in this area in order to get a clearer picture.

*Continental Drift.*

Finally we come to the interpretation of results in terms of continental drift of the Indian landmass during the period of eruption of Deccan traps. Several workers have given thought to this problem. Previous estimates of the drift of the Indian landmass are based upon the calculation of mean directions for the Lower and the Upper Deccan traps studied from various localities. These estimates are of the order of 26°-27° in Latitude (DEUTSCH *et al.*, 1958; BLACKETT *et al.*, 1960). MCELHINNY (1968), however has stated that there has been a very small amount of drift (of the order of 7° in latitude) between the corresponding periods. The difference in the two interpretations is mainly in the calculation of mean pole positions for the time interval covering the lower and the upper Deccan traps. MCELHINNY (1968) has included all published data from various localities and has assumed that the scatter in the directions from all normal as well as reverse sites is entirely due to palaeosecular variation during the normal and the reverse periods. This assumption appears to be questionable.

We have calculated the pole positions for vertical sequences adequately sampled (at least five flows over 800 ft. thick) and studied in detail. We have assumed that the mean directions of each vertical sequence (Table 2) could be taken as representative of sufficient time during which secular variation has been averaged. We can therefore get reliable estimate of the possible amount of continental drift that has taken place during the period of eruption of lower and upper Deccan traps discussed above. The maximum amount of drift is shown by the difference in pole positions between Khandala and Jabalpur sequences. This difference being of the order of 23° in latitude it is safe to conclude that appreciable drift has taken place between the period of Lower and Upper Deccan trap eruption and the earlier estimate of drift by DEUTSCH *et al.* (1958) is of the right order of magnitude.

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## References

BHIMASANKARAM, V. L. S. and PAL, P. C., 1968, *Paleomagnetic polarity transition (?) during Deccan trap volcanism.* Jour. of Osmania Univ. (Science), Golden Jubilee Vol., p. 43-61.

BLACKETT, P. M. S., CLEGG, J. A., STUBBS, P. H. S., 1960, *An analysis of rock magnetic data.* Proc. Roy. Soc., Ser. A, Vol. 256, p. 291-322.

CLEGG, J. A., DEUTSCH, E. R. and GRIFFITHS, D. H., 1956, *Rock magnetism in India.* Phil. Mag. (8), 1, p. 419-431.

CREER, K. M., 1959, *A. C. demagnetization of unstable Keuper Marls from S. W. England.* Geophys. J., 2, p. 261.

————, 1965, *Application of rock magnetism to investigations of the secular variation during geologic times.* In '*Magnetism and Cosmos*'. Proc. of Nato advanced study Inst. on Planetary and Stellar Magnetism, Oliver and Boyd, Edinbunrgh, p. 45-48.

DEUTSCH, E. R., RADHAKRISHNAMURTHY, C. and SAHASRABUDHE, P. W., 1958, *The remanent magnetism of some lavas in the Deccan traps.* Phil. Mag., 3, p. 170-184.

————, ———— and ————, 1959, *Paleomagnetism of the Deccan traps.* Ann. Geophys. (Paris) 15, p. 39-59.

KRISHNAN, M. S., 1968, *Geology of India and Burma.* Higginbothams, Madras, India, p. 405-408.

MCELHINNY, M. W., 1968, *Northward drift of India - Examination of recent Paleomagnetic results.* Nature, 217, p. 342-344.

SAHASRABUDHE, P. W., 1963, *Paleomagnetism and Geology of the Deccan traps.* Proc. of Seminar on « Geophysical Investigations of Peninsular Shield », Indian Geophysical Union, Hyderabad. India, p. 226-243.

SIGURGEIRSSON, TH., 1957, *Direction of magnetization in Icelandic basalts.* Phil. Mag., Adv. Phys. 6, p. 240-246.

VERMA, R. K. and APPA RAO, A., 1963, *Paleomagnetism of Deccan traps from Chincholi, Vikarabad and Gulbarga areas.* Proc. of Seminar on « Geophysical Investigations in the Peninsular Shield », Indian Geophysical Union, Hyderabad, India, p. 161-168.

ZIJL, J. S. VAN, GRAHAM, K. W. T. and HALES, A. L., 1962, *The paleomagnetism of the Stromberg lavas from South Afriea.* Geophys. J., 6, p. 169-182.